

Discrete-Time Signal Processing

Exam Problem Set

Full statements + prerequisite definitions

Based on the uploaded revised lecture notes (Sections 1–8) and
Oppenheim, Schafer & Buck, *Discrete-Time Signal Processing*, 2nd ed.

Scope. This packet targets the material covered before the FIR/IIR section of the revised notes: discrete-time sequences, signal classes, sampling/aliasing/rate conversion, system properties, LTI systems, convolution, correlation, and inverse systems. Problems or subparts whose solution substantially relies on DTFT/frequency-response machinery not developed in the revised notes are explicitly marked as optional or omitted.

Prepared for exam review

Problem statements are reproduced from the user-provided textbook scan. Definitions are distilled from the user-provided revised notes, with a few explicitly labeled textbook-only definitions needed for optional problems.

Contents

1	Notation and definitions you need	2
2	Reference diagrams for the sampling problems	7
2.1	Figure 4.11: $C/D \rightarrow$ discrete-time processing $\rightarrow D/C$	7
2.2	Figure 4.20: sampling-rate reduction by an integer factor	7
2.3	Figure 4.28: rational sampling-rate conversion by L/M	8
3	Problem set	9
3.1	A. Sequences, indexing, and periodicity	9
3.2	B. System properties: linearity, time invariance, causality, memory, stability	11
3.3	C. LTI systems, convolution, impulse responses, and inverses	17
3.4	D. Correlation	23
3.5	E. Sampling, aliasing, and rate conversion	24
3.6	F. Quantization supplement	29
4	Suggested solving order	30

Notation and definitions you need

Discrete-time sequence

A discrete-time sequence is a function

$$x : \mathbb{Z} \rightarrow \mathbb{C}, \quad n \mapsto x[n].$$

Its *support* is $\text{supp } x = \{n \in \mathbb{Z} : x[n] \neq 0\}$. A sequence is finite-support if this set is finite; right-sided if its support is bounded below; left-sided if its support is bounded above.

Unit impulse, unit step, shifts, reversal, and conjugation

$$\delta[n] = \begin{cases} 1, & n = 0, \\ 0, & n \neq 0, \end{cases} \quad u[n] = \begin{cases} 1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

The delay/shift operator is

$$(S_r x)[n] = x[n - r].$$

Reversal is $(Rx)[n] = x[-n]$. Complex conjugation is pointwise: $x^*[n] = \overline{x[n]}$. The shifted impulse $S_k \delta = \delta[n - k]$ is the standard basis sequence with its single 1 at index k . Every sequence has the pointwise impulse expansion

$$x[n] = \sum_{k \in \mathbb{Z}} x[k] \delta[n - k].$$

Periodicity

A positive integer N is a period of x if $x[n + N] = x[n]$ for every n . The least positive period is the *fundamental period*. A nonzero complex exponential $Ae^{j(\omega_0 n + \phi)}$ is periodic exactly when $\omega_0/(2\pi) \in \mathbb{Q}$. If $\omega_0/(2\pi) = p/q$ in lowest terms, the fundamental period is q .

Boundedness, absolute summability, energy, and power

$$\|x\|_\infty = \sup_n |x[n]|, \quad \|x\|_1 = \sum_n |x[n]|, \quad E_x = \|x\|_2^2 = \sum_n |x[n]|^2.$$

The spaces ℓ^∞ , ℓ^1 , and ℓ^2 consist of sequences for which these quantities are finite, respectively. When the limit exists, symmetric average power is

$$P_x = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K |x[n]|^2.$$

For sequences on $\mathbb{Z}$, $\ell^1 \subsetneq \ell^2 \subsetneq \ell^\infty$.

System and the notation $T\{x\}[n]$

A discrete-time system is a map $T : \mathcal{X} \rightarrow \mathcal{Y}$ between specified classes of sequences. If $y = T\{x\}$, then

$$T\{x\}[n] = y[n]$$

means the value at index n of the entire output sequence produced when the entire input sequence is x .

Memoryless, linear, time-invariant, causal

A system is *memoryless* if its output at n depends only on $x[n]$; equivalently $T\{x\}[n] = F_n(x[n])$ for some functions F_n .

It is *linear* if, for all admissible inputs and scalars,

$$T\{ax_1 + bx_2\} = aT\{x_1\} + bT\{x_2\}.$$

It is *time-invariant* if it commutes with every integer shift:

$$TS_r = S_rT.$$

It is *causal* if agreement of two inputs through time N forces equality of their outputs at time N (equivalently, through time N).

BIBO stability

A system is bounded-input bounded-output (BIBO) stable if every admissible bounded input produces a bounded output:

$$\|x\|_\infty < \infty \implies \|Tx\|_\infty < \infty.$$

One bounded input producing an unbounded response is enough to disprove BIBO stability.

Impulse response and step response

Whenever the relevant inputs are in the domain,

$$h[n] = T\{\delta\}[n], \quad s[n] = T\{u\}[n].$$

These are the system's impulse and step responses. Their definitions alone do not assume linearity or time invariance.

LTI system and convolution

An LTI system is both linear and time-invariant. For a fixed sequence h , the convolution system is

$$T_h\{x\}[n] = (x * h)[n] = \sum_{k \in \mathbb{Z}} x[k]h[n - k],$$

whenever the sum is defined. Convolution by a fixed h is LTI. Conversely, an LTI system has this form on finite-support inputs with $h = T\{\delta\}$, and on more general inputs when the required convergence/continuity conditions hold.

Key LTI tests

For a convolution system:

- **Causality:** T_h is causal iff $h[n] = 0$ for every $n < 0$.
- **Memoryless:** T_h is memoryless iff $h[n] = c\delta[n]$ for some constant c .
- **BIBO stability:** T_h is BIBO stable iff

$$\sum_{n \in \mathbb{Z}} |h[n]| < \infty,$$

i.e. $h \in \ell^1$.

Cascade and parallel interconnections of convolution systems

If two convolution systems have responses h_1, h_2 , then a cascade is equivalent to one convolution system with

$$h_{\text{cascade}} = h_1 * h_2,$$

while parallel branches whose outputs are added are equivalent to

$$h_{\text{parallel}} = h_1 + h_2.$$

Inverse system and convolution inverse

A system $T : \mathcal{X} \rightarrow \mathcal{Y}$ is invertible on $\mathcal{X}$ if there exists $G : T(\mathcal{X}) \rightarrow \mathcal{X}$ such that

$$G(Tx) = x \quad \forall x \in \mathcal{X}.$$

An inverse exists on the range exactly when T is injective. For convolution systems T_h and T_g , exact inverse cascading means

$$T_g(T_h x) = x,$$

and the corresponding condition is

$$h * g = \delta,$$

subject to the relevant domain/convergence assumptions.

Accumulator and first backward difference

On a left-finite class of inputs,

$$(Ax)[n] = \sum_{k \leq n} x[k], \quad (\nabla x)[n] = x[n] - x[n-1].$$

The accumulator has impulse response $u[n]$; the first difference has impulse response $\delta[n] - \delta[n-1]$. On the stated left-finite class, $\nabla A = A\nabla = I$.

Correlation

For finite-energy sequences $x, y \in \ell^2$, the revised notes use

$$r_{xy}[\ell] = \sum_{k \in \mathbb{Z}} x[k] y^*[k - \ell].$$

Autocorrelation is r_{xx} . If $y_e[n] = y^*[-n]$, then

$$r_{xy} = x * y_e.$$

For real sequences, conjugation does nothing.

Important convention warning for Problem 2.59: the textbook writes its autocorrelation as

$$R_x[n] = \sum_{k=-\infty}^{\infty} x^*[k] x[n+k],$$

which is the same idea with a different lag convention. Use the convention stated in the problem you are solving.

Uniform sampling and frequency variables

Sampling a continuous-time signal $x_c(t)$ with period T gives

$$x[n] = x_c(nT), \quad f_s = \frac{1}{T}.$$

If f is analog frequency in hertz and $\Omega = 2\pi f$ is analog angular frequency in rad/s, then the corresponding digital frequency is

$$\omega = \Omega T = \frac{2\pi f}{f_s} \text{ rad/sample.}$$

Digital frequencies differing by $2\pi k$ describe the same complex-exponential sequence.

Spectrum / spectral content (notation used in the sampling problems)

For a continuous-time signal $x_c(t)$, the textbook writes its Fourier transform or *spectrum* as $X_c(j\Omega)$. When the ordinary Fourier integral exists,

$$X_c(j\Omega) = \int_{-\infty}^{\infty} x_c(t) e^{-j\Omega t} dt.$$

Informally, the signal's *spectral content* is the set of frequencies at which this frequency-domain representation is nonzero. Thus saying that a baseband signal is band-limited to Ω_M means $X_c(j\Omega) = 0$ for $|\Omega| > \Omega_M$; in hertz, $f_{\max} = \Omega_M/(2\pi)$.

Frequency response of an LTI system (textbook notation)

For an LTI system with impulse response $h[n]$, its frequency response is

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n},$$

whenever this sum is defined. It describes the complex gain applied to a complex-exponential input: if $x[n] = e^{j\omega n}$, then the corresponding LTI output is $H(e^{j\omega})e^{j\omega n}$. You need this terminology mainly to read a few textbook statements; most selected pre-FIR/IIR questions can be solved directly in the time domain.

Aliasing, band limitation, and Nyquist terminology

Aliasing is the failure of distinct admissible continuous-time signals to remain distinguishable after sampling. Sampled complex exponentials alias when their analog frequencies differ by an integer multiple of f_s ; zero-phase real cosines also have the $f \leftrightarrow -f$ symmetry.

A baseband signal is band-limited to $f_{\max}$ if its spectrum vanishes outside $[-f_{\max}, f_{\max}]$. Its Nyquist rate is $2f_{\max}$. For a chosen sampling rate f_s , the Nyquist frequency is $f_s/2$. The robust sampling condition used in the notes is

$$f_s > 2f_{\max}.$$

Upsampling, downsampling, and rational rate conversion

For integers $L, M \geq 1$,

$$(U_L x)[n] = \begin{cases} x[n/L], & L \mid n, \\ 0, & L \nmid n, \end{cases} \quad (D_M x)[n] = x[Mn].$$

Upsampling inserts $L - 1$ zeros between successive samples. Downsampling keeps every M th sample.

Rational rate conversion by L/M is implemented by upsampling, low-pass filtering, then downsampling.

Ideal C/D and D/C blocks in the textbook sampling problems

C/D denotes ideal continuous-to-discrete conversion (sampling); D/C denotes ideal discrete-to-continuous reconstruction of the corresponding band-limited continuous-time signal. In the textbook's Figure 4.11, a continuous-time input is sampled, processed by a discrete-time system, then ideally reconstructed.

Ideal discrete-time low-pass filter

An ideal low-pass filter with cutoff ω_c has frequency response

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| < \omega_c, \\ 0, & \omega_c < |\omega| \leq \pi, \end{cases}$$

with its 2π -periodic extension understood. This definition is needed only for a few sampling questions that explicitly mention such a filter.

Uniform quantization and the high-resolution noise model

For input range $[-A, A]$ and B bits, there are $L = 2^B$ levels and step size

$$\Delta = \frac{2A}{L}.$$

For a mid-rise quantizer, an in-range sample has quantization error $e_q = x_q - x$ satisfying $|e_q| \leq \Delta/2$. In the standard high-resolution approximation,

$$e_q \sim \text{Unif}[-\Delta/2, \Delta/2], \quad \mathbb{E}[e_q] = 0, \quad \mathbb{E}[e_q^2] = \frac{\Delta^2}{12}.$$

The quantization-noise question at the end of this packet is marked optional because it also uses random-process terminology beyond the main revised-note development.

Optional random-process terminology for Problem 4.57

For the purposes of this optional problem, a discrete-time random process is *wide-sense stationary* if its mean is constant and its autocorrelation depends only on the lag, not on absolute time. A zero-mean stationary white-noise sequence with variance σ_x^2 satisfies

$$\mathbb{E}\{x[n]\} = 0, \quad \Phi_{xx}[m] = \mathbb{E}\{x[n]x^*[n+m]\} = \sigma_x^2\delta[m].$$

Two sequences are *uncorrelated* at a pair of sample times when their cross-covariance there is zero. The signal-to-quantizing-noise ratio (SQNR) requested in Problem 4.57 is the ratio of input-signal variance to quantization-error variance.

Reference diagrams for the sampling problems

These figures are included because several selected Chapter 4 questions refer to them by number.

Figure 4.11: C/D $\rightarrow$ discrete-time processing $\rightarrow$ D/C

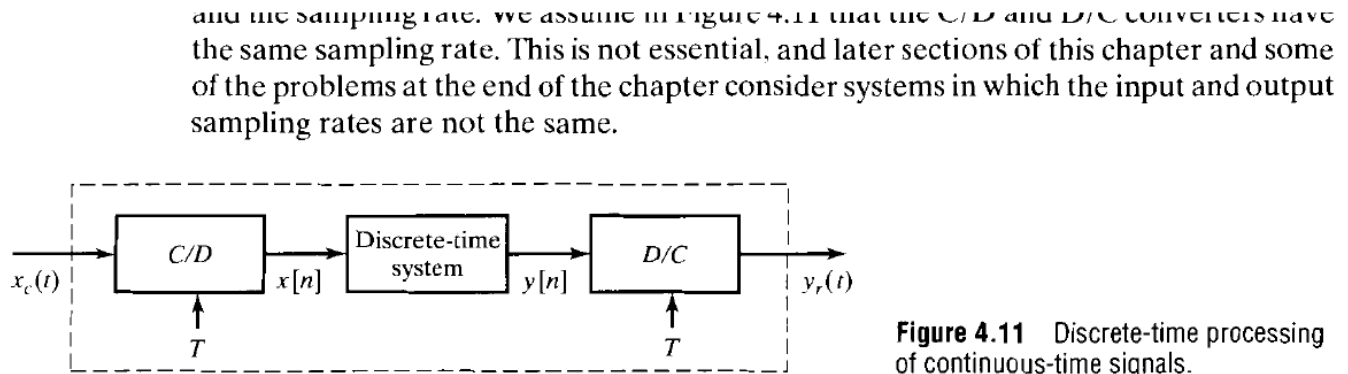


Figure 4.20: sampling-rate reduction by an integer factor

Equation (4.71) defines the system depicted in Figure 4.20, which is called a *sampling rate compressor* (see Crochiere and Rabiner, 1983) or simply a *compressor*. From Eq. (4.71), it is clear that $x_d[n]$ is identical to the sequence that would be obtained from $x_c(t)$ by

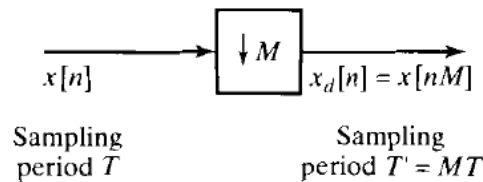
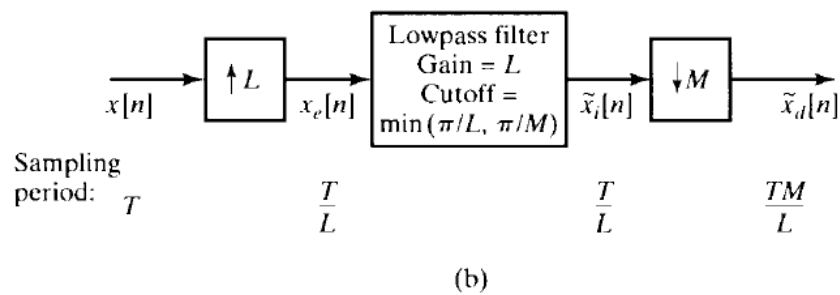
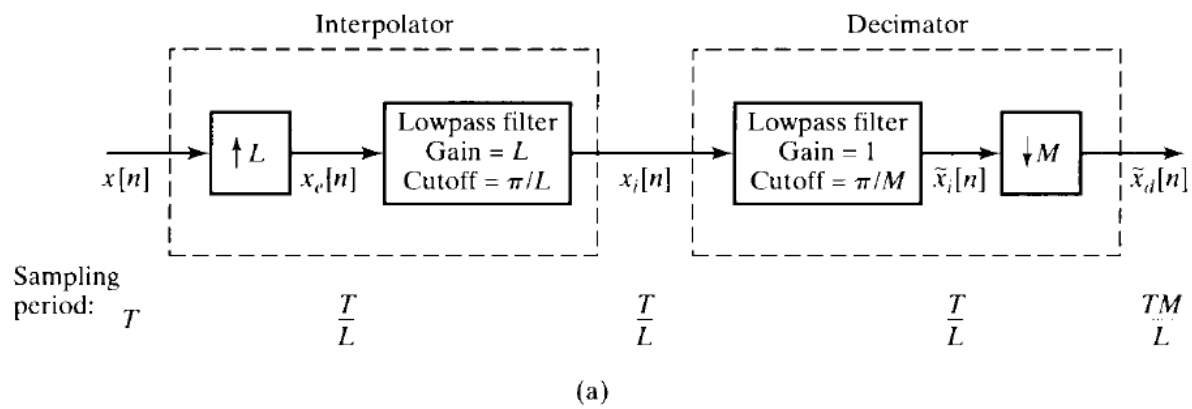


Figure 4.20 Representation of a compressor or discrete-time sampler.

Figure 4.28: rational sampling-rate conversion by L/M

Sec. 4.6 Changing the Sampling Rate Using Discrete-Time Processing

177



Problem set

Priority key. **[CORE]** means I would do it before the exam even under time pressure. **[ADDITIONAL]** means worthwhile once the core set is comfortable. **[STRETCH/OPTIONAL]** means useful but it leans on frequency-domain/random-process facts not developed as fully in the revised notes.

A. Sequences, indexing, and periodicity

Problems 2.7 [CORE]

$$1 + \frac{1}{2}e^{-j\omega} + \frac{1}{4}e^{-j2\omega}$$

2.7. Determine whether each of the following signals is periodic. If the signal is periodic, state its period.

- (a) $x[n] = e^{j(\pi n/6)}$
- (b) $x[n] = e^{j(3\pi n/4)}$
- (c) $x[n] = [\sin(\pi n/5)]/(\pi n)$
- (d) $x[n] = e^{j\pi n/\sqrt{2}}$

Problems 2.28–2.30 [CORE]

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DISCRETE-TIME SIGNALS AND SYSTEMS Chap. 2

2.28. Determine which of the following signals is periodic. If a signal is periodic, determine its period.

(a) $x[n] = e^{j(2\pi n/5)}$

(b) $x[n] = \sin(\pi n/19)$

(c) $x[n] = ne^{j\pi n}$

(d) $x[n] = e^{jn}$

2.29. A discrete-time signal $x[n]$ is shown in Figure P2.29-1.



Figure P2.29-1

Sketch and label carefully each of the following signals:

(a) $x[n - 2]$

(b) $x[4 - n]$

(c) $x[2n]$

(d) $x[n]u[2 - n]$

(e) $x[n - 1]\delta[n - 3]$

2.30. For each of the following systems, determine whether the system is (1) stable, (2) causal, (3) linear, and (4) time invariant.

(a) $T(x[n]) = (\cos \pi n)x[n]$

(b) $T(x[n]) = x[n^2]$

(c) $T(x[n]) = x[n] \sum_{k=0}^{\infty} \delta[n - k]$

(d) $T(x[n]) = \sum_{k=n-1}^{\infty} x[k]$

2.31. Consider the difference equation

B. System properties: linearity, time invariance, causality, memory, stability

Problem 2.1 [CORE]

- 2.1.** For each of the following systems, determine whether the system is (1) stable, (2) causal, (3) linear, (4) time invariant, and (5) memoryless:

- (a) $T(x[n]) = g[n]x[n]$ with $g[n]$ given
- (b) $T(x[n]) = \sum_{k=n_0}^n x[k]$
- (c) $T(x[n]) = \sum_{k=n-n_0}^{n+n_0} x[k]$
- (d) $T(x[n]) = x[n - n_0]$
- (e) $T(x[n]) = e^{x[n]}$
- (f) $T(x[n]) = ax[n] + b$
- (g) $T(x[n]) = x[-n]$
- (h) $T(x[n]) = x[n] + 3u[n + 1]$

Problem 2.14 [ADDITIONAL]

(f) $(1/4)^n u[n] + 4^n u[-n - 1]$

- 2.14.** A single input–output relationship is given for each of the following three systems:

- (a) System A: $x[n] = (1/3)^n$, $y[n] = 2(1/3)^n$.
- (b) System B: $x[n] = (1/2)^n$, $y[n] = (1/4)^n$.
- (c) System C: $x[n] = (2/3)^n u[n]$, $y[n] = 4(2/3)^n u[n] - 3(1/2)^n u[n]$.

Based on this information, pick the strongest possible conclusion that you can make about each system from the following list of statements:

- (i) The system cannot possibly be LTI.
- (ii) The system must be LTI.
- (iii) The system can be LTI, and there is only one LTI system that satisfies this input–output constraint.
- (iv) The system can be LTI, but cannot be uniquely determined from the information in this input–output constraint.

If you chose option (iii) from this list, specify either the impulse response $h[n]$ or the frequency response $H(e^{j\omega})$ for the LTI system.

Problem 2.15 [ADDITIONAL]

- 2.15.** Consider the system illustrated in Figure P2.15-1. The output of an LTI system with an impulse response $h[n] = \left(\frac{1}{4}\right)^n u[n + 10]$ is multiplied by a unit step function $u[n]$ to yield the output of the overall system. Answer each of the following questions, and briefly justify your answers:

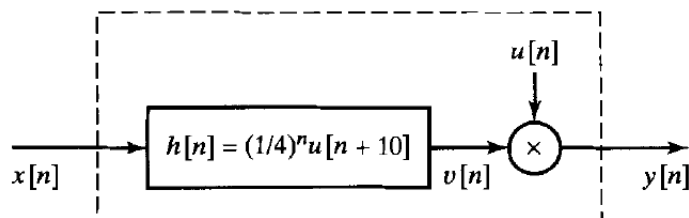


Figure P2.15-1

- (a) Is the overall system LTI?
- (b) Is the overall system causal?
- (c) Is the overall system stable in the BIBO sense?

2.16. Consider the following difference equation:

Problems 2.21 and 2.23 [CORE]

In the reproduced block below, Problem 2.22 is also present because it is itself a core convolution problem; do all three.

- (b) For what range of values of a will the system be stable?

Basic Problems

- 2.21. Consider an arbitrary linear system with input $x[n]$ and output $y[n]$. Show that if $x[n] = 0$ for all n , then $y[n]$ must also be zero for all n .
- 2.22. For each of the pairs of sequences in Figure P2.22-1, use discrete convolution to find the response to the input $x[n]$ of the linear time-invariant system with impulse response $h[n]$.

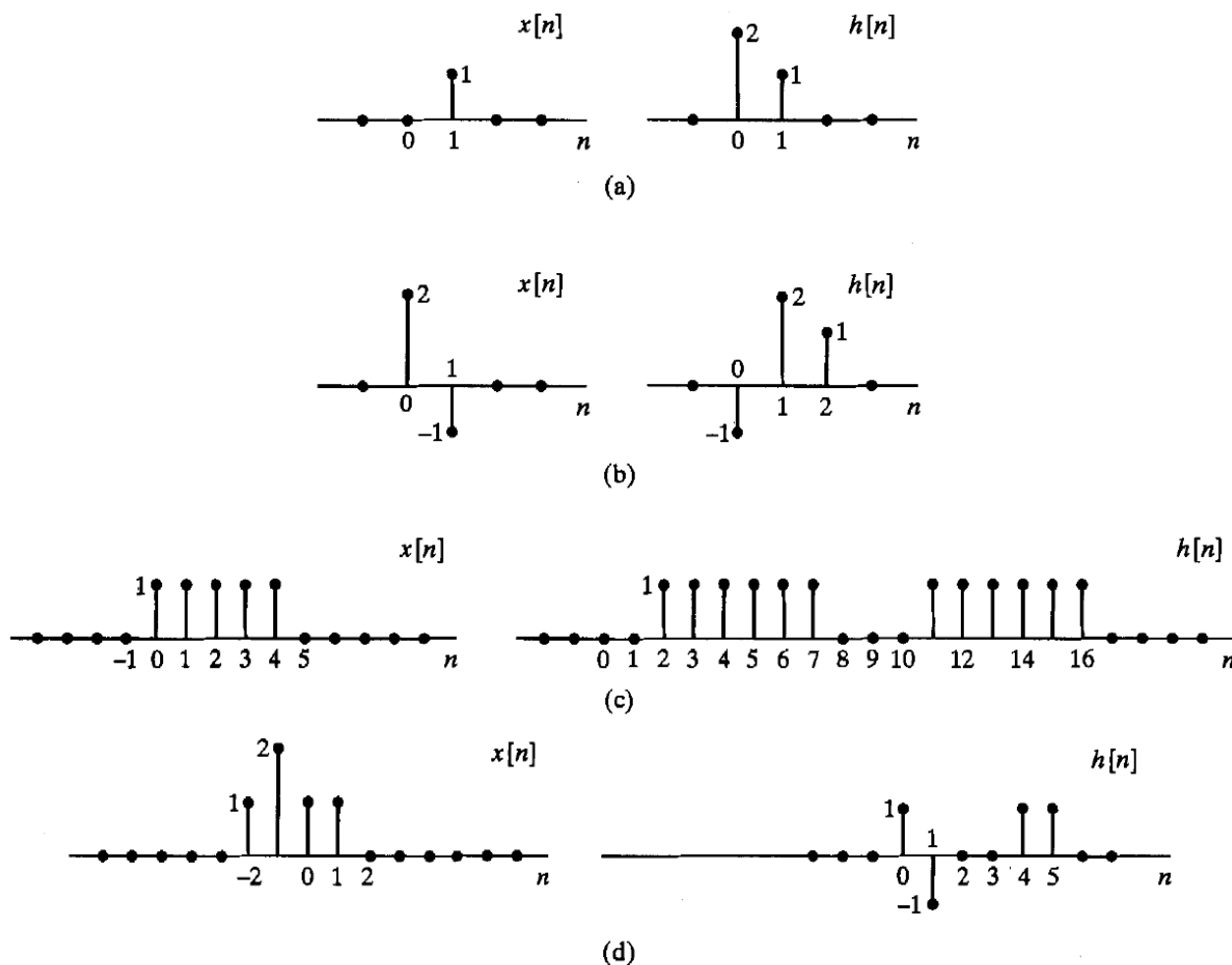


Figure P2.22-1

- 2.23. Using the definition of linearity (Eqs. (2.26a)–(2.26b)), show that the ideal delay system (Example 2.3) and the moving-average system (Example 2.4) are both linear systems.

Problem 2.35 [ADDITIONAL]

Advanced Problems

2.35. The system T in Figure P2.35-1 is known to be *time invariant*. When the inputs to the system are $x_1[n]$, $x_2[n]$, and $x_3[n]$, the responses of the system are $y_1[n]$, $y_2[n]$, and $y_3[n]$, as shown.

78

Discrete-Time Signals and Systems Chap. 2

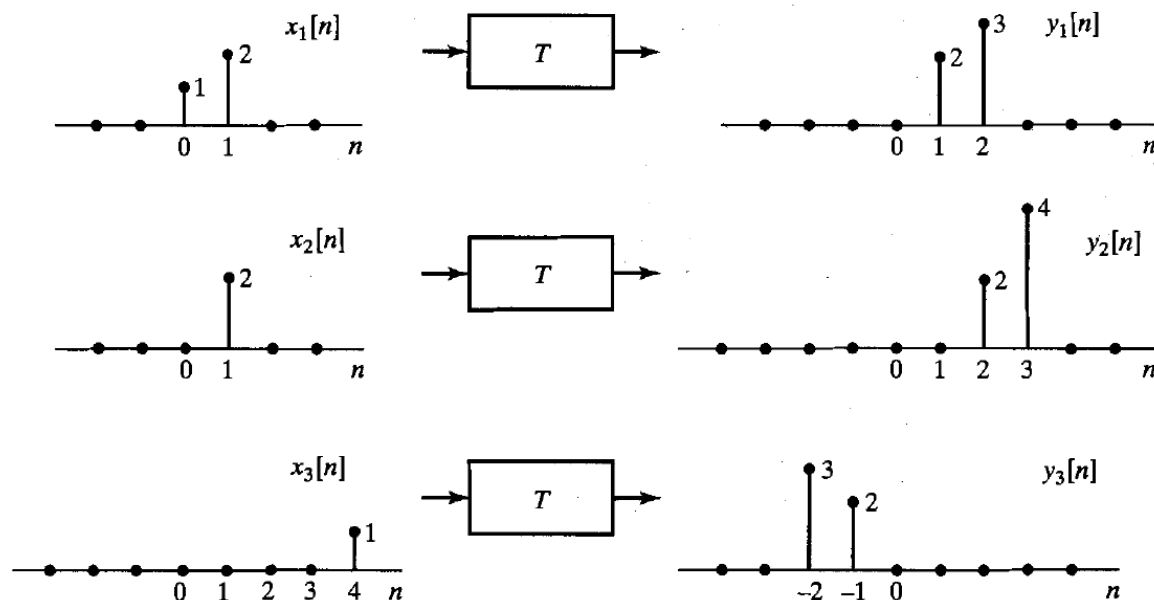


Figure P2.35-1

- Determine whether the system T could be linear.
- If the input $x[n]$ to the system T is $\delta[n]$, what is the system response $y[n]$?
- What are all possible inputs $x[n]$ for which the response of the system T can be determined from the given information alone?

Problem 2.36 [ADDITIONAL]

2.36. The system L in Figure P2.36-1 is known to be *linear*. Shown are three output signals $y_1[n]$, $y_2[n]$, and $y_3[n]$ in response to the input signals $x_1[n]$, $x_2[n]$, and $x_3[n]$, respectively.

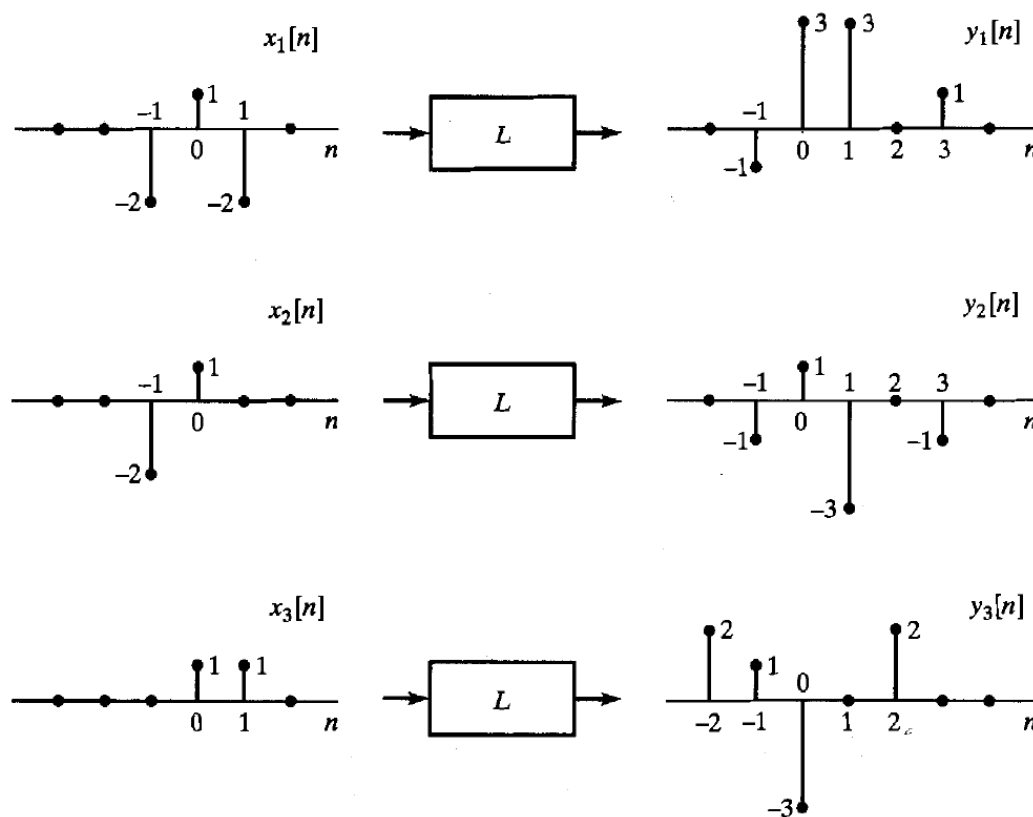


Figure P2.36-1

Chap. 2 PROBLEMS

13

- Determine whether the system L could be time invariant.
- If the input $x[n]$ to the system L is $\delta[n]$, what is the system response $y[n]$?

Problem 2.37 [CORE]

2.37. Consider a discrete-time linear time-invariant system with impulse response $h[n]$. If the input $x[n]$ is a periodic sequence with period N (i.e., if $x[n] = x[n + N]$), show that the output $y[n]$ is also a periodic sequence with period N .

Problem 2.62 [CORE]

- 2.62.** The causality of a system was defined in Section 2.2.4. From this definition, show that, for a linear time-invariant system, causality implies that the impulse response $h[n]$ is zero for $n < 0$. One approach is to show that if $h[n]$ is *not* zero for $n < 0$, then the system *cannot*

86

Discrete-Time Signals and Systems Chap. 2

be causal. Show also that if the impulse response is zero for $n < 0$, then the system will necessarily be causal.

Problem 2.68 [ADDITIONAL]

- 2.68.** Figure P1.68-1 shows the input–output relationships of Systems A and B, while Figure P1.68-2 contains two possible cascade combinations of these systems.

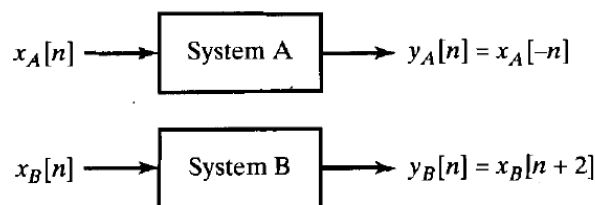


Figure P2.68-1

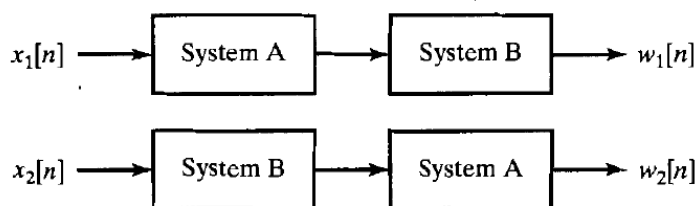


Figure P2.68-2

88

Discrete-Time Signals and Systems Chap. 2

If $x_1[n] = x_2[n]$, will $w_1[n]$ and $w_2[n]$ necessarily be equal? If your answer is *yes*, clearly and concisely explain why and demonstrate with an example. If your answer is *not necessarily*, demonstrate with a counterexample.

C. LTI systems, convolution, impulse responses, and inverses

Problems 2.2–2.3 [CORE]

- 2.2. (a)** The impulse response $h[n]$ of a linear time-invariant system is known to be zero, except in the interval $N_0 \leq n \leq N_1$. The input $x[n]$ is known to be zero, except in the interval $N_2 \leq n \leq N_3$. As a result, the output is constrained to be zero, except in some interval $N_4 \leq n \leq N_5$. Determine N_4 and N_5 in terms of N_0 , N_1 , N_2 , and N_3 .
- (b)** If $x[n]$ is zero, except for N consecutive points, and $h[n]$ is zero, except for M consecutive points, what is the maximum number of consecutive points for which $y[n]$ can be nonzero?
- 2.3.** By direct evaluation of the convolution sum, determine the step response of a linear time-invariant system whose impulse response is

$$h[n] = a^{-n}u[-n], \quad 0 < a < 1.$$

Problem 2.10 [CORE]

- LTI, but that has $y[0] = y[1] = 1$. Find the response of this system to $x[n] = \delta[n]$.
- 2.10.** Determine the output of a linear time-invariant system if the impulse response $h[n]$ and the input $x[n]$ are as follows:
- (a)** $x[n] = u[n]$ and $h[n] = a^n u[-n - 1]$, with $a > 1$.
 - (b)** $x[n] = u[n - 4]$ and $h[n] = 2^n u[-n - 1]$.
 - (c)** $x[n] = u[n]$ and $h[n] = (0.5)2^n u[-n]$.
 - (d)** $h[n] = 2^n u[-n - 1]$ and $x[n] = u[n] - u[n - 10]$
- Use your knowledge of linearity and time invariance to minimize the work in Parts (b)–(d).
- 2.11.** Consider an LTI system with frequency response

Problems 2.18–2.19 [CORE]

- (c) Sketch the magnitude of $R(e^{j\omega})$ and $\tilde{W}(e^{j\omega})$ for the case when $M = 4$.
- 2.18.** For each of the following impulse responses of LTI systems, indicate whether or not the system is causal:
- (a) $h[n] = (1/2)^n u[n]$
 - (b) $h[n] = (1/2)^n u[n - 1]$
 - (c) $h[n] = (1/2)^{|n|}$
 - (d) $h[n] = u[n + 2] - u[n - 2]$
 - (e) $h[n] = (1/3)^n u[n] + 3^n u[-n - 1]$
- 2.19.** For each of the following impulse responses of LTI systems, indicate whether or not the system is stable:
- (a) $h[n] = 4^n u[n]$
 - (b) $h[n] = u[n] - u[n - 10]$
 - (c) $h[n] = 3^n u[-n - 1]$

- (d) $h[n] = \sin(\pi n/3) u[n]$
- (e) $h[n] = (3/4)^{|n|} \cos(\pi n/4 + \pi/4)$
- (f) $h[n] = 2u[n + 5] - u[n] - u[n - 5]$

Problem 2.22 [CORE]

Problem 2.22 appears in the block reproduced earlier with Problems 2.21 and 2.23. It is intentionally not duplicated here.

Problems 2.24–2.25 [CORE]

Chap. 2 Problems

75

- 2.24.** The impulse response of a linear time-invariant system is shown in Figure P2.24-1. Determine and carefully sketch the response of this system to the input $x[n] = u[n - 4]$.

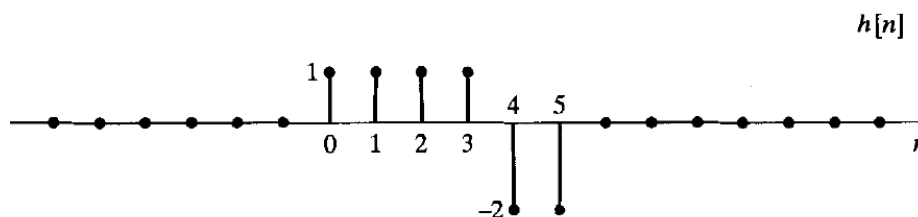


Figure P2.24-1

- 2.25.** A linear time-invariant system has impulse response $h[n] = u[n]$. Determine the response of this system to the input $x[n]$ shown in Figure P2.25-1 and described as

$$x[n] = \begin{cases} 0, & n < 0, \\ a^n, & 0 \leq n \leq N_1, \\ 0, & N_1 < n < N_2, \\ a^{n-N_2}, & N_2 \leq n \leq N_2 + N_1, \\ 0, & N_2 + N_1 < n, \end{cases}$$

 where $0 < a < 1$.


Problem 2.34 [CORE]

Figure P2.34-1

2.34. The input–output pair shown in Figure P2.34-1 is given for a stable LTI system.

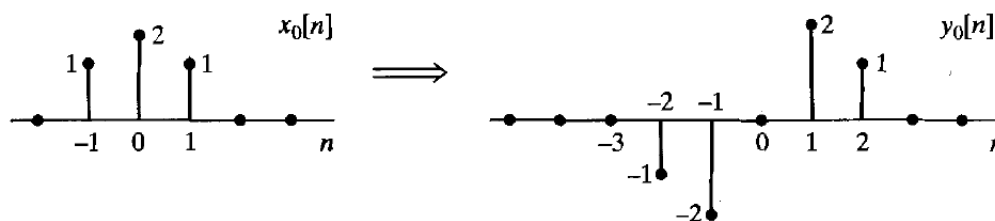


Figure P2.34-1

(a) Determine the response to the input $x_1[n]$ in Figure P2.34-2.

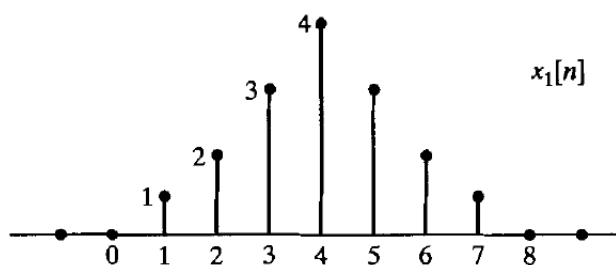


Figure P2.34-2

(b) Determine the impulse response of the system.

Problem 2.42, parts (a) and (d) [ADDITIONAL]

Do parts (a) and (d). Parts (b)–(c) rely on frequency-response/difference-equation machinery outside the main pre-FIR/IIR packet.

$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n + 16k].$$

Find the output of the system.

2.42. Consider the system in Figure P2.42-1.

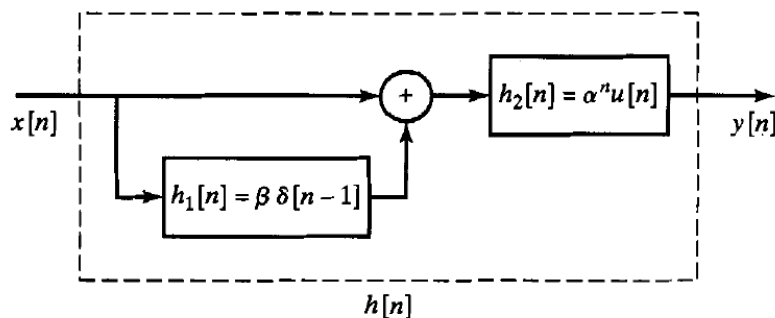


Figure P2.42-1

- (a) Find the impulse response $h[n]$ of the overall system.
- (b) Find the frequency response of the overall system.
- (c) Specify a difference equation that relates the output $y[n]$ to the input $x[n]$.
- (d) Is this system causal? Under what condition would the system be stable?

Problem 2.60 [CORE]

- (b) Show that the Fourier transform of $R_x[n]$ is equal to $|X(e^{j\omega})|^2$.

2.60. The signals $x[n]$ and $y[n]$ shown in Figure P2.60-1 are the input and corresponding output for an LTI system.

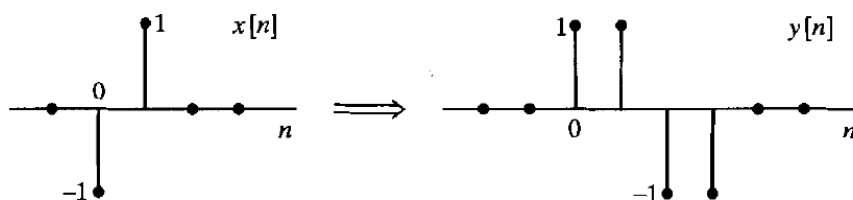


Figure P2.60-1

- (a) Find the response of the system to the sequence $x_2[n]$ in Figure P2.60-2.

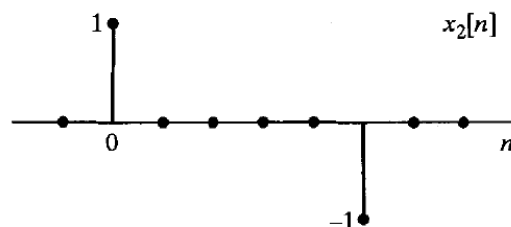


Figure P2.60-2

- (b) Find the impulse response $h[n]$ for this LTI system.

2.61 Consider a system for which the input $x[n]$ and output $y[n]$ satisfy the difference equation

Problem 2.70, parts (a), (b), (d), and (e) [CORE]

Skip part (c) unless your instructor expects frequency-response calculations.

$$H_2(e^{j\omega}) = \begin{cases} \text{unspecified,} & |\omega| \leq 0.4\pi, \\ 0, & 0.4\pi < |\omega| \leq \pi. \end{cases}$$

Suppose also that the input $x[n]$ is known to be bandlimited to 0.3π , i.e.,

$$X(e^{j\omega}) = \begin{cases} \text{unspecified,} & |\omega| < 0.3\pi, \\ 0, & 0.3\pi \leq |\omega| \leq \pi. \end{cases}$$

Over what region of $-\pi \leq \omega < \pi$ is $Y(e^{j\omega})$, the DTFT of $y[n]$, guaranteed to be zero?

2.70. A commonly used numerical operation called the *first backward difference* is defined as

$$y[n] = \nabla(x[n]) = x[n] - x[n-1],$$

where $x[n]$ is the input and $y[n]$ is the output of the first-backward-difference system.

- (a) Show that this system is linear and time invariant.
- (b) Find the impulse response of the system.
- (c) Find and sketch the frequency response (magnitude and phase).
- (d) Show that if

$$x[n] = f[n] * g[n],$$

then

$$\nabla(x[n]) = \nabla(f[n]) * g[n] = f[n] * \nabla(g[n]),$$

where $*$ denotes discrete convolution.

- (e) Find the impulse response of a system that could be cascaded with the first-difference system to recover the input; i.e., find $h_i[n]$, where

D. Correlation

Problem 2.59, part (a) [CORE]

Do part (a). Part (b) is a Fourier-transform statement and is outside the main revised-note scope.

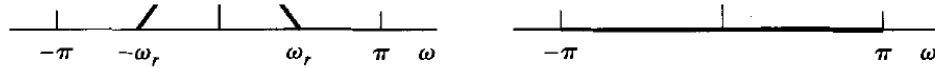


Figure P2.58-2

2.59. The autocorrelation sequence of a signal $x[n]$ is defined as

$$R_x[n] = \sum_{k=-\infty}^{\infty} x^*[k]x[n+k].$$

- (a) Show that for an appropriate choice of the signal $g[n]$, $R_x[n] = x[n] * g[n]$, and identify the proper choice for $g[n]$.

E. Sampling, aliasing, and rate conversion

Problems 4.1–4.4 [CORE]

PROBLEMS**Basic Problems with Answers**

4.1. The signal

$$x_c(t) = \sin(2\pi(100)t)$$

was sampled with sampling period $T = 1/400$ second to obtain a discrete-time signal $x[n]$. What is the resulting signal $x[n]$?

4.2. The sequence

$$x[n] = \cos\left(\frac{\pi}{4}n\right), \quad -\infty < n < \infty.$$

was obtained by sampling a continuous-time signal

$$x_c(t) = \cos(\Omega_0 t), \quad -\infty < t < \infty,$$

at a sampling rate of 1000 samples/s. What are two possible positive values of Ω_0 that could have resulted in the sequence $x[n]$?

4.3. The continuous-time signal

$$x_c(t) = \cos(4000\pi t)$$

is sampled with a sampling period T to obtain a discrete-time signal

$$x[n] = \cos\left(\frac{\pi n}{3}\right).$$

- (a) Determine a choice for T consistent with this information.
- (b) Is your choice for T in Part (a) unique? If so, explain why. If not, specify another choice of T consistent with the information given.

4.4. The continuous-time signal

$$x_c(t) = \sin(20\pi t) + \cos(40\pi t)$$

is sampled with a sampling period T to obtain the discrete-time signal

$$x[n] = \sin\left(\frac{\pi n}{5}\right) + \cos\left(\frac{2\pi n}{5}\right).$$

- (a) Determine a choice for T consistent with this information.
- (b) Is your choice for T in Part (a) unique? If so, explain why. If not, specify another choice

Problem 4.5(a) [ADDITIONAL]

Use the Figure 4.11 reference diagram given earlier. Do part (a).

- (a) Determine a choice for T consistent with this information.
 - (b) Is your choice for T in Part (a) unique? If so, explain why. If not, specify another choice of T consistent with the information given.
- 4.5.** Consider the system of Figure 4.11, with the discrete-time system an ideal lowpass filter with cutoff frequency $\pi/8$ radians/s.
- (a) If $x_c(t)$ is bandlimited to 5 kHz, what is the maximum value of T that will avoid aliasing in the C/D converter?
 - (b) If $1/T = 10$ kHz, what will the cutoff frequency of the effective continuous-time filter be?
 - (c) Repeat Part (b) for $1/T = 20$ kHz.

Problems 4.10–4.11 [CORE]

- (b) Find the value of $x[5]$. (*Hint: Find values for all odd-indexed points.*)
 - (c) Let $y[n]$ be the decimated version of $x[n]$, i.e., $y[n] = x[2n]$. Can you reconstruct $x[n]$ from $y[n]$ for all n . If yes, how? If no, justify your answer.
- 4.10.** Each of the following continuous-time signals is used as the input $x_c(t)$ for an ideal C/D converter as shown in Figure 4.1 with the sampling period T specified. In each case, find the resulting discrete-time signal $x[n]$.
- (a) $x_c(t) = \cos(2\pi(1000)t)$, $T = (1/3000)$ sec
 - (b) $x_c(t) = \sin(2\pi(1000)t)$, $T = (1/1500)$ sec
 - (c) $x_c(t) = \sin(2\pi(1000)t)/(\pi t)$, $T = (1/5000)$ sec
- 4.11.** The following continuous-time input signals $x_c(t)$ and corresponding discrete-time output signals $x[n]$ are those of an ideal C/D as shown in Figure 4.1. Specify a choice for the sampling period T that is consistent with each pair of $x_c(t)$ and $x[n]$. In addition, indicate whether your choice of T is unique. If not, specify a second possible choice of T consistent with the information given.
- (a) $x_c(t) = \sin(10\pi t)$, $x[n] = \sin(\pi n/4)$
 - (b) $x_c(t) = \sin(10\pi t)/(10\pi t)$, $x[n] = \sin(\pi n/2)/(\pi n/2)$
- 4.12.** In the system of Figure 4.11, assume that

Problem 4.14 [STRETCH/OPTIONAL]

This is a good multirate/aliasing exercise, but parts involving sinc-type sequences use spectral facts more explicitly than the revised notes develop.

- $y[n]$.
- (c) In general, how does the continuous-time LTI system $h_c(t)$ limit the range of the sampling period T that can be used without changing $y[n]$?
- 4.14.** Which of the following signals can be downsampled by a factor of 2 using the system in Figure 4.20 without any loss of information?
- (a) $x[n] = \delta[n - n_0]$, for n_0 some unknown integer
- (b) $x[n] = \cos(\pi n/4)$
- (c) $x[n] = \cos(\pi n/4) + \cos(3\pi n/4)$

Problems 4.16–4.17 [STRETCH/OPTIONAL]

Use Figure 4.28 given earlier. These are useful rate-conversion exercises but lean on spectral intuition for some inputs.

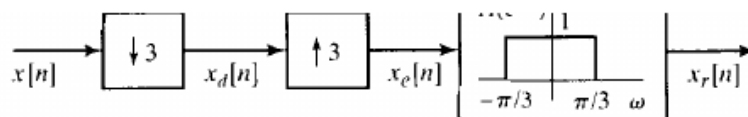


Figure P4.15-1

- 4.16.** Consider the system in Figure 4.28. The input $x[n]$ and corresponding output $\tilde{x}_d[n]$ are given for a specific choice of M/L in each of the following parts. Determine a choice for M/L based on the information given, and specify whether your choice is unique.
- (a) $x[n] = \sin(\pi n/3)/(\pi n/3)$, $\tilde{x}_d[n] = \sin(5\pi n/6)/(5\pi n/6)$
- (b) $x[n] = \cos(3\pi n/4)$, $\tilde{x}_d[n] = \cos(\pi n/2)$
- 4.17.** Each of the following parts lists an input signal $x[n]$ and the upsampling and downsampling rates L and M for the system in Figure 4.28. Determine the corresponding output $\tilde{x}_d[n]$.
- (a) $x[n] = \sin(2\pi n/3)/\pi n$, $L = 4$, $M = 3$
- (b) $x[n] = \sin(3\pi n/4)$, $L = 3$, $M = 5$
- 4.18.** For the system shown in Figure 4.28, $X(e^{j\omega})$, the Fourier transform of the input signal

Problem 4.20(a) [CORE]

Do part (a). Part (b) requires the ideal discrete-time low-pass filter's passband interpretation and can be attempted if that was emphasized in class.

$$\frac{1}{T} \quad \frac{1}{T}$$

Figure P4.19-2

- 4.20.** Consider the system in Figure 4.11. The input signal $x_c(t)$ has the Fourier transform shown in Figure P4.20-1 with $\Omega_0 = 2\pi(1000)$ radians/second. The discrete-time system is an ideal lowpass filter with frequency response

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| < \omega_c, \\ 0, & \text{otherwise.} \end{cases}$$

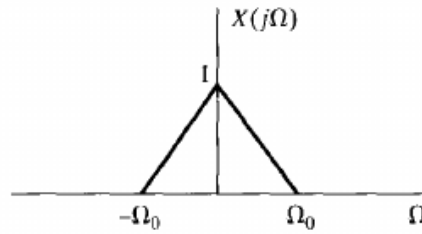


Figure P4.20-1

- (a) What is the minimum sampling rate $F_s \approx 1/T$ such that no aliasing occurs in sampling the input?
- (b) If $\omega_c = \pi/2$, what is the minimum sampling rate such that $y_c(t) = x_c(t)$?

— — — —

Problem 4.23 [ADDITIONAL]

- 4.23.** In Figure P4.23-1, assume that $X_c(j\Omega) = 0$, $|\Omega| \geq \pi/T_1$. For the general case in which $T_1 \neq T_2$ in the system, express $y_c(t)$ in terms of $x_c(t)$. Is the basic relationship different for $T_1 > T_2$ and $T_1 < T_2$?

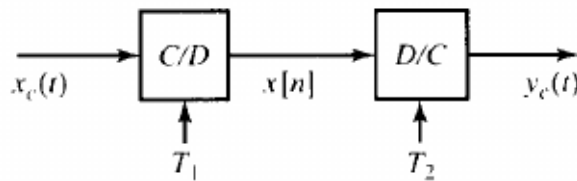


Figure P4.23-1

Problem 4.32(a) [CORE]

Do part (a). Parts (b)–(c) are frequency-response design questions beyond the main revised-note scope.

Sampling of Continuous-Time Signals

Chap. 4

4.32. A bandlimited continuous-time signal is known to contain a 60-Hz component, which we want to remove by processing with the system of Figure 4.11, where $T = 10^{-4}$.

(a) What is the highest frequency that the continuous-time signal can contain if aliasing is to be avoided?

(b) The discrete-time system to be used has frequency response

$$H(e^{j\omega}) = \frac{[1 - e^{-j(\omega - \omega_0)}][1 - e^{-j(\omega + \omega_0)}]}{[1 - 0.9e^{-j(\omega - \omega_0)}][1 - 0.9e^{-j(\omega + \omega_0)}]}.$$

Sketch the magnitude and phase of $H(e^{j\omega})$.

(c) What value should be chosen for ω_0 to eliminate the 60-Hz component?

4.33. Consider the system in Figure 4.11 with $X_c(j\Omega) = 0$ for $|\Omega| \geq 2\pi(1000)$ and the discrete-time system a squarer, i.e., $y[n] = x^2[n]$. What is the largest value of T such that $y_c(t) =$

F. Quantization supplement

Problem 4.57(a)–(b) [STRETCH/OPTIONAL]

Optional. It connects directly to the revised notes' high-resolution quantization-noise model, but it also uses stationary-random-process language. Do parts (a)–(b); part (c) is a filtering/noise-power extension.

Figure 4.48-2

- 4.57.** As discussed in Section 4.8.2, to process sequences on a digital computer, we must quantize the amplitude of the sequence to a set of discrete levels. This quantization can be expressed in terms of passing the input sequence $x[n]$ through a quantizer $Q(x)$ that has an input-output relation as shown in Figure 4.48.

As discussed in Section 4.8.3, if the quantization interval Δ is small compared with changes in the level of the input sequence, we can assume that the output of the quantizer is of the form

$$y[n] = x[n] + e[n],$$

where $e[n] = Q(x[n]) - x[n]$ and $e[n]$ is a stationary random process with a first-order probability density uniformly distributed between $-\Delta/2$ and $\Delta/2$, uncorrelated from sample to sample and uncorrelated with $x[n]$, so that $\mathcal{E}\{e[n]x[m]\} = 0$ for all m and n .

Let $x[n]$ be a stationary white-noise process with zero mean and variance σ_x^2 .

- (a) Find the mean, variance, and autocorrelation sequence of $e[n]$.
- (b) What is the signal-to-quantizing-noise ratio σ_x^2/σ_e^2 ?
- (c) The quantized signal $y[n]$ is to be filtered by a digital filter with impulse response $h[n] = \frac{1}{2}[a^n + (-a)^n]u[n]$. Determine the variance of the noise produced at the output due to the input quantization noise, and determine the signal-to-noise ratio at the output.

Suggested solving order

If time is tight, I would work in this order:

1. 2.1, 2.7, 2.28–2.30: basic sequence manipulations and system classification.
2. 2.2–2.3, 2.10, 2.18–2.19, 2.22, 2.24–2.25: convolution and impulse-response fundamentals.
3. 2.34, 2.37, 2.60, 2.62, 2.70(a,b,d,e): deeper LTI reasoning, causality, and inverses.
4. 2.59(a): correlation-as-convolution.
5. 4.1–4.4, 4.10–4.11, 4.20(a), 4.32(a): sampling and aliasing essentials.
6. 2.14, 2.15, 2.35–2.36, 2.42(a,d), 2.68, 4.5(a), 4.23: additional conceptual depth.
7. 4.14, 4.16–4.17, 4.57(a,b): stretch/optional material.

Exam-readiness test. You should be able to state the definitions at the front of this packet without looking them up, prove the LTI causality and BIBO criteria at the level of the revised notes, compute finite convolutions reliably, classify unfamiliar systems, reason about aliasing from $\omega = \Omega T$, and explain why an inverse may depend on the admissible signal class.